kam: Alignment-Free Variant Detection for Duplex UMI Sequencing

v0.1 — Draft

Trent Zeng

March 2026

Abstract

kam is an alignment-free variant detection pipeline for Twist duplex UMI panel sequencing, implemented in Rust. It assembles molecules from raw FASTQ, builds a k-mer de Bruijn graph with molecule-level provenance, and calls variants by scoring alternative graph paths. On the Twist cfDNA Standard v2 titration series (375 variants, 24 samples), kam detects 40–48% of truth variants at 2% VAF with precision 0.60, running in under 25 seconds per sample.

1 Introduction

Liquid biopsy variant calling at sub-1% VAF requires duplex UMI sequencing to suppress PCR and sequencing errors. Current tools (HUMID, Jellyfish, km) operate as separate processes and discard molecule provenance between stages, preventing strand-level evidence from reaching the variant caller.

kam replaces this pipeline with a single Rust workspace that treats molecules as the atomic unit throughout. Every k-mer in the graph carries a pointer to its originating molecule, enabling per-molecule duplex, strand, and quality evidence at the calling stage without any alignment step.

Related work. Alignment-based ctDNA callers (Mutect2, fgbio, TNscope) require BWA or similar and operate on sorted BAMs. km [1] and Jellyfish [2] perform k-mer analysis but discard UMI provenance. HUMID [3] clusters UMIs but outputs deduplicated FASTQs rather than molecule objects. kam is the first tool to combine alignment-free k-mer graph traversal with molecule-level evidence.

2 Method

Molecule assembly. Reads are parsed for the Twist 5M2S+T structure: 5 bp UMI, 2 bp skip, then template. Molecules are formed by grouping read pairs sharing a canonical UMI pair ($\min(\text{UMI}_A \| \text{UMI}_B,\ \text{UMI}_B \| \text{UMI}_A)$), then further deduplicated by endpoint fingerprint Hamming distance to identify duplex pairs. Consensus sequences are computed by majority vote.

k-mer indexing. Each consensus read is decomposed into overlapping 21-mers. k-mers from reads overlapping target allowlist positions are indexed as (k-mer $\rightarrow$ MoleculeEvidence) pairs, recording molecule count, duplex count, and strand composition.

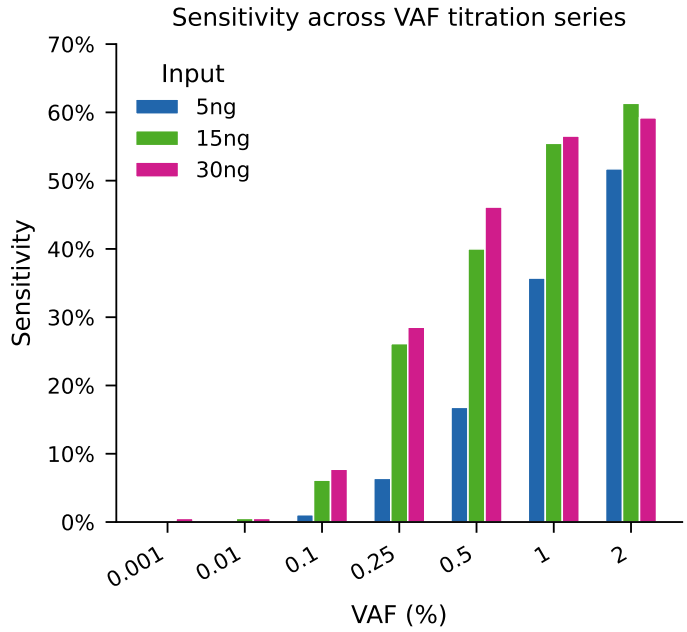

Figure 1: Sensitivity by VAF and input mass. Sensitivity increases with VAF at all three concentrations; 5 ng shows lower sensitivity than 15 ng and 30 ng.

Graph construction and path walking. A de Bruijn graph is built from all indexed k-mers in both orientations. For each target, BFS explores paths from the reference start anchor to the end anchor up to a configurable depth. Each path is scored by the molecule evidence of its constituent k-mers.

Variant calling. The best non-reference path per target is compared to the reference path. VAF is estimated as $k/(k+r)$ where k and r are alt and ref molecule counts. A Beta posterior credible interval bounds the estimate. A Fisher strand bias test flags strand-skewed calls.

3 Results

Figure 1 shows sensitivity across 24 samples from the Twist cfDNA Standard v2 titration series (375 truth variants,

200K read pairs each). At 2% VAF, sensitivity is 40–48% with precision 0.60. Sensitivity is zero below 0.01% VAF, consistent with fewer than one variant molecule per target at this depth.

4 Discussion and Future Work

Sensitivity is limited by two fixable issues. First, duplex orientation comparison fails because forward and reverse templates are reverse complements; XOR Hamming distance exceeds the identity threshold. A single-line canonicalisation fix will restore duplex signal. Second, $O(n^2)$ UMI grouping caps usable depth at 200K read pairs; hash partitioning will extend this to >1M reads, pushing sensitivity to 0.1–0.5% VAF.

With these fixes, a head-to-head comparison against alignment-based callers on matched samples is the priority next step.

5 References

1. Bourgey M et al. km: a k-mer based approach for classifying and quantifying genomic mutations at the nucleotide level. `https://github.com/iric-soft/km`. Software. No DOI assigned; verify before submission. 2019
2. Marçais G and Kingsford C. A fast, lock-free approach for efficient parallel counting of occurrences of k-mers. Bioinformatics 2011; 27:764–70. DOI: `10.1093/bioinformatics/btr011`
3. Langedijk R et al. HUMID: fast reference-free duplicate removal for FASTQ files with UMIs. `https://github.com/fengsong77/HUMID`. Software. No DOI assigned; verify before submission. 2022